

Built-in Sorting Functions of Arrays in PHP

Sorting means arranging those items in a particular order:

- **Ascending order** → Small to big, A to Z
- **Descending order** → Big to small, Z to A

PHP gives us many built-in sorting functions to arrange arrays easily.

1. `sort()` — Sort values in ascending order

Use when: You only care about values and want them from smallest to largest.

```
<?php
$numbers = [5, 2, 8, 1, 3];

sort($numbers);

print_r($numbers);
?>
```

Output:

```
Array
(
    [0] => 1
    [1] => 2
    [2] => 3
    [3] => 5
    [4] => 8
)
```

Important:

- Re-indexes the array starting from 0
- Removes old keys

Example:

```
$colors = [
    10 => "Red",
    5  => "Blue",
    20 => "Green"
];

sort($colors);

print_r($colors);
```

Output:

```
Array
(
    [0] => Blue
    [1] => Green
    [2] => Red
)
```

2. `rsort()` — Sort values in descending order

R = Reverse

```
<?php
$numbers = [5, 2, 8, 1, 3];

rsort($numbers);

print_r($numbers);
?>
```

Output:

```
Array
(
    [0] => 8
    [1] => 5
    [2] => 3
    [3] => 2
    [4] => 1
)
```

3. `asort()` — Sort values but keep keys

A = Associative

Useful when keys have meaning.

Example:

```
<?php
$marks = [
    "John" => 80,
    "Alex" => 95,
    "Sam" => 70
];

asort($marks);

print_r($marks);
?>
```

Output:

```
Array
(
    [Sam] => 70
    [John] => 80
    [Alex] => 95
)
```

Notice:

- Values are sorted
- Names (keys) stay attached

4. `arsort()` — Sort values descending and keep keys

```
<?php
$marks = [
    "John" => 80,
    "Alex" => 95,
    "Sam" => 70
];

arsort($marks);

print_r($marks);
?>
```

Output:

```
Array
(
    [Alex] => 95
    [John] => 80
    [Sam] => 70
)
```

5. `ksort()` — Sort by keys ascending

K = Key

```
<?php
$students = [
    "John" => 80,
    "Alex" => 95,
    "Sam" => 70
];

ksort($students);

print_r($students);
?>
```

Output:

```
Array
(
    [Alex] => 95
    [John] => 80
    [Sam] => 70
)
```

Keys are sorted alphabetically.

6. `krsort()` — Sort by keys descending

```
<?php
$students = [
    "John" => 80,
    "Alex" => 95,
    "Sam" => 70
];

krsort($students);

print_r($students);
?>
```

Output:

```
Array
(
    [Sam] => 70
    [John] => 80
    [Alex] => 95
)
```

Quick Memory Trick

Function	Sorts By	Direction	Keeps Keys?
<code>sort()</code>	Values	Ascending	No
<code>rsort()</code>	Values	Descending	No
<code>asort()</code>	Values	Ascending	Yes
<code>arsort()</code>	Values	Descending	Yes
<code>ksort()</code>	Keys	Ascending	Yes
<code>krsort()</code>	Keys	Descending	Yes

Simple Rule to Remember

- **Need values sorted?** → `sort()` / `rsort()`
- **Need keys to stay attached?** → `asort()` / `arsort()`
- **Need keys sorted?** → `ksort()` / `krsort()`

These functions cover most array sorting needs in PHP.